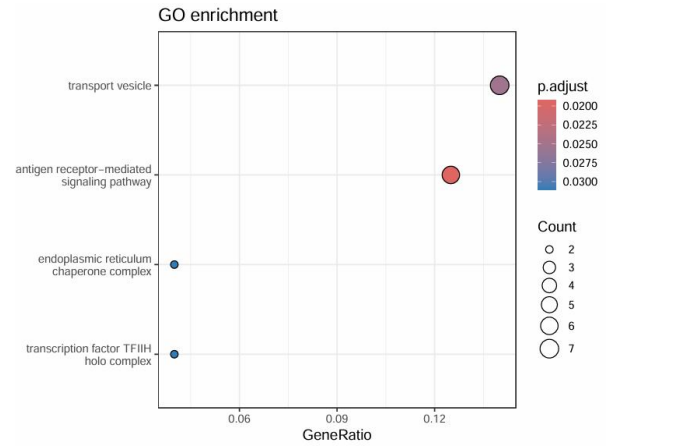
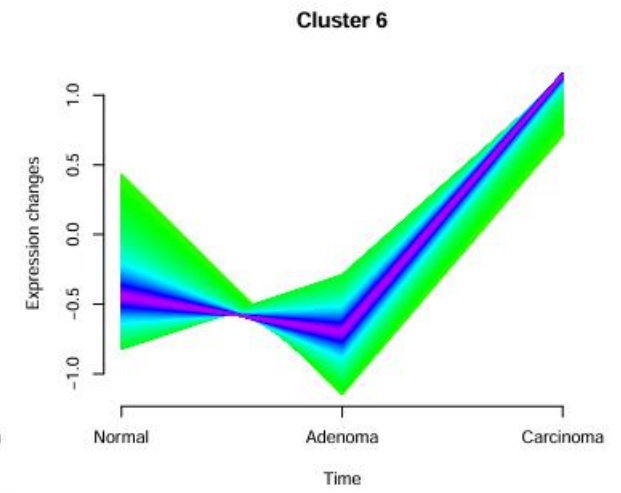
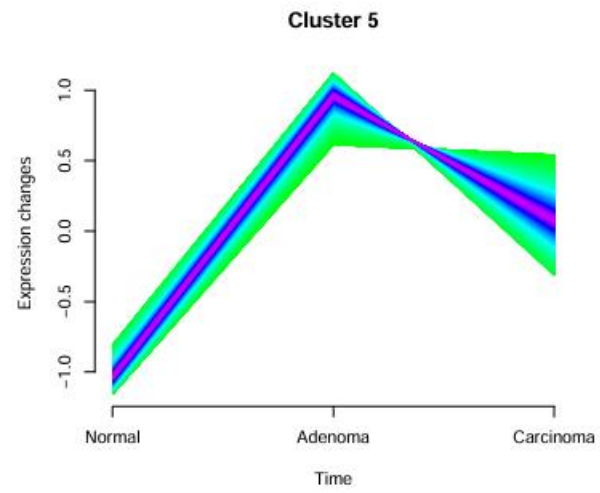
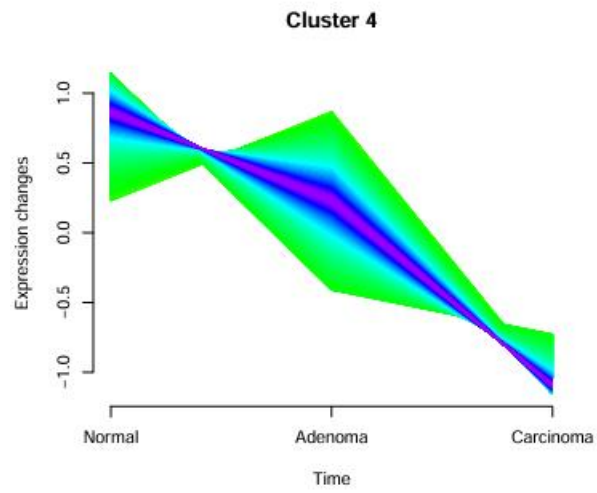
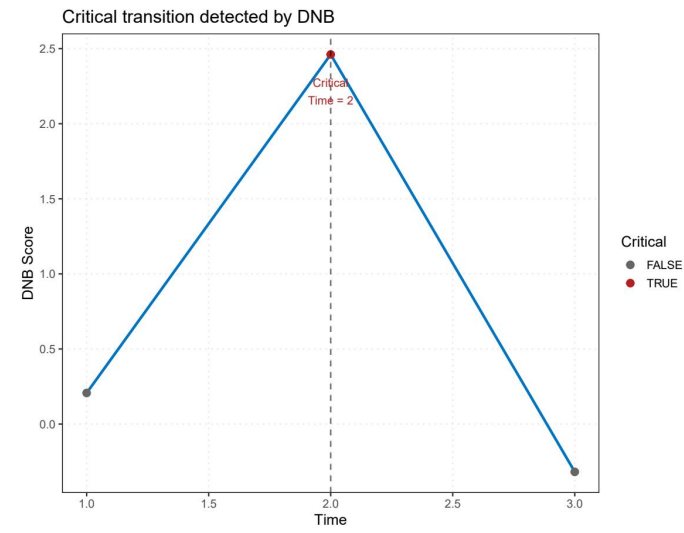
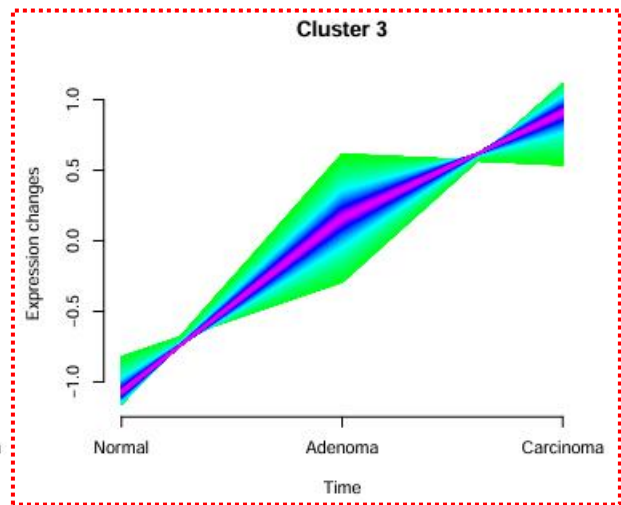
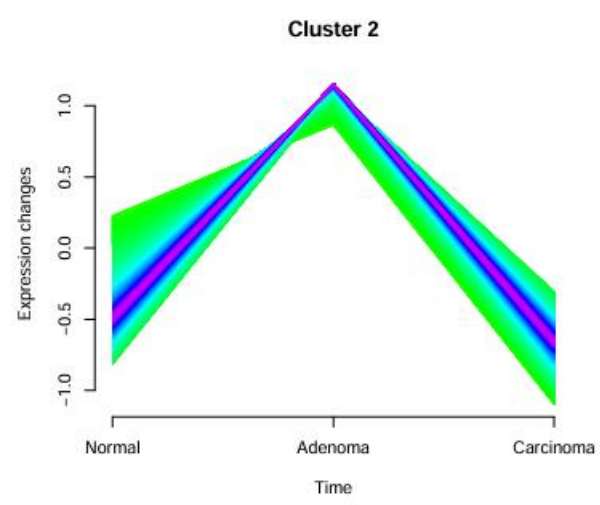
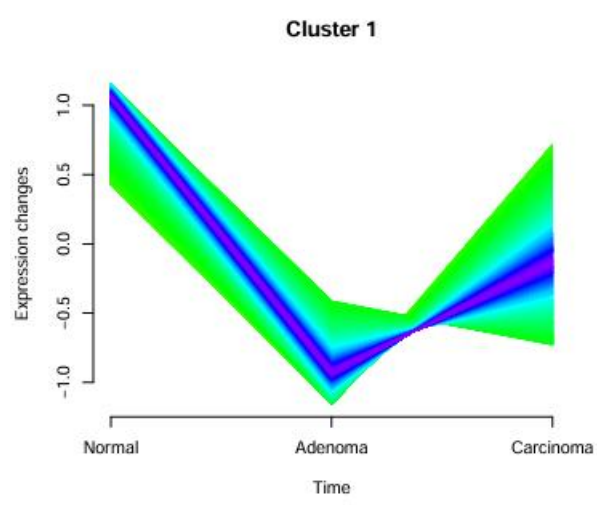
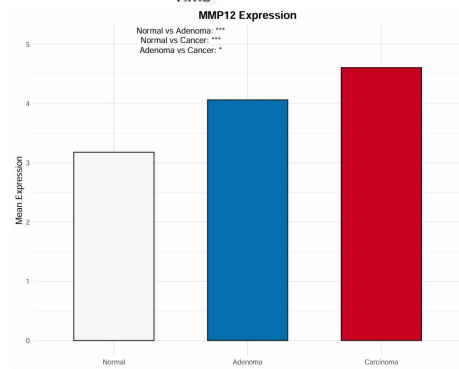


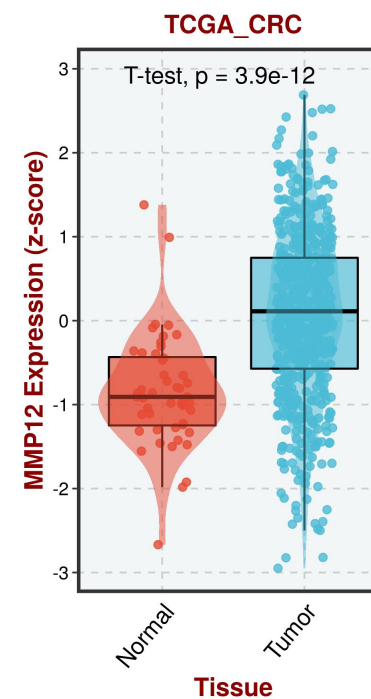
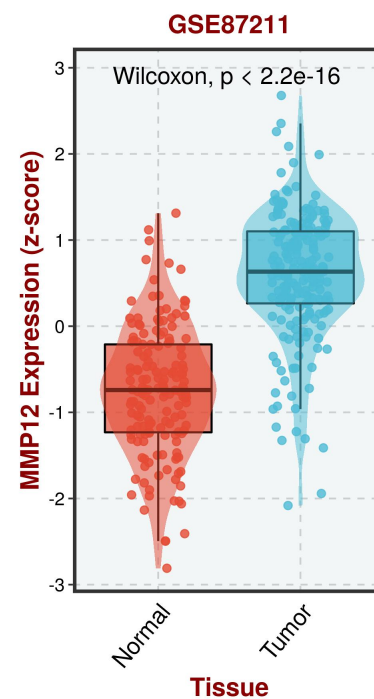
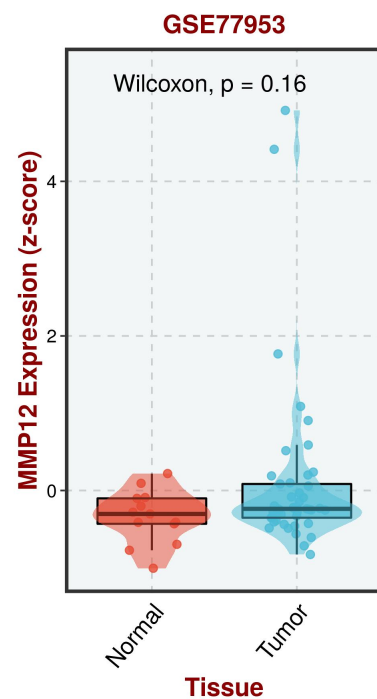
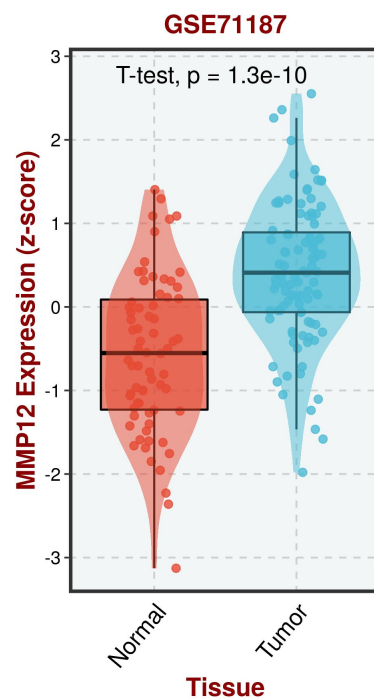
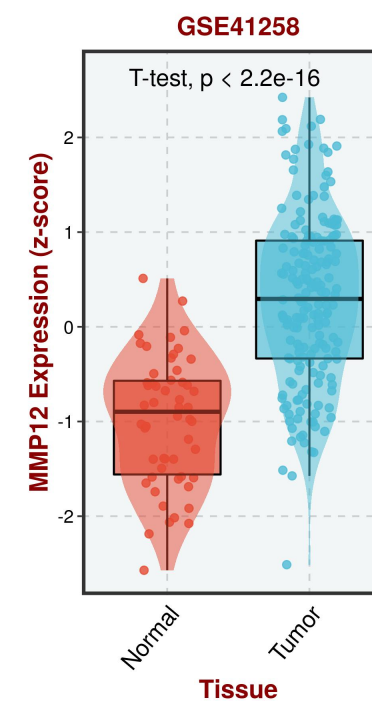
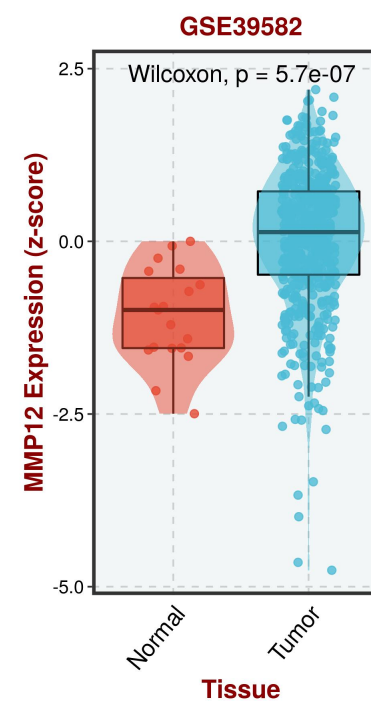
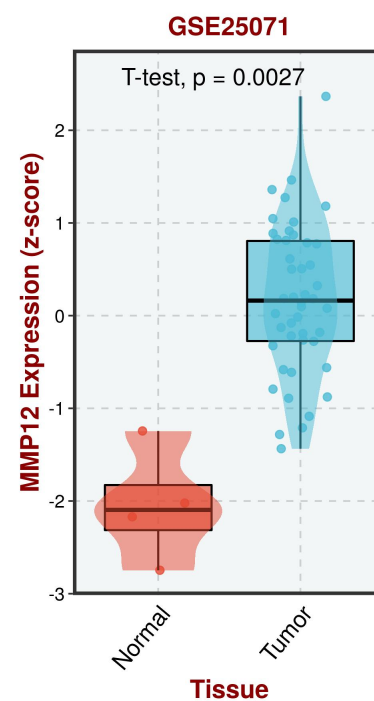
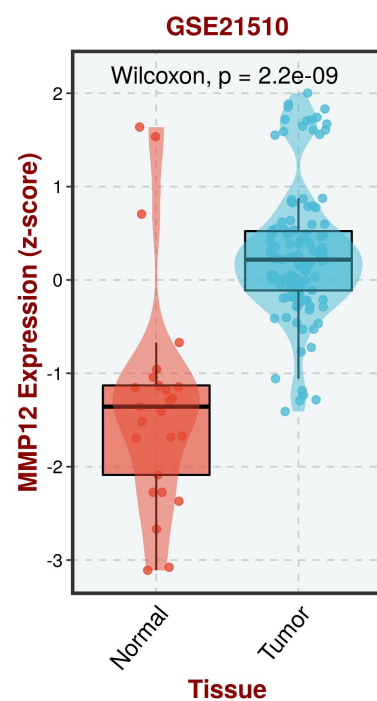
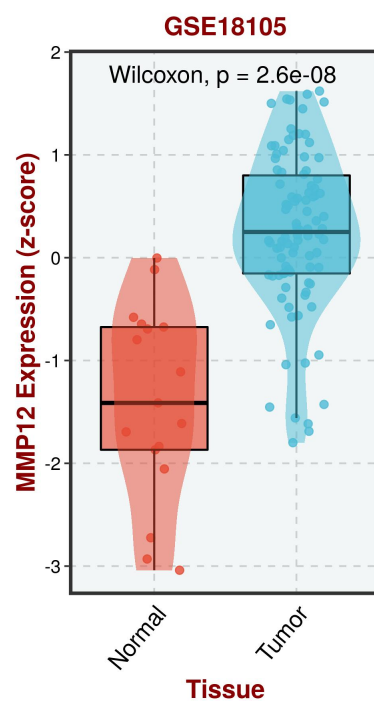
- 1.需要验证 正常细胞系和癌细胞系的MMP12表达水平
- 2.需要构建MMP12敲低过表达质粒
- 3.总结各类型MMP12的文章
- 4.文章大概率免疫方面做，后面分析发现和免疫很相关
- 5.做MMP12双重作用，高表达促进炎癌转变，癌症进展又会导致MMP12降低

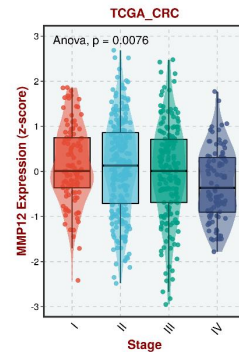
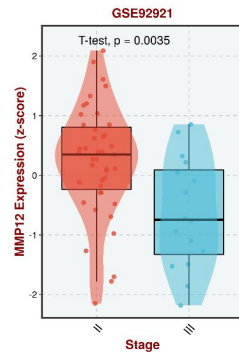
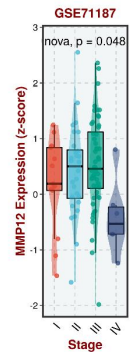
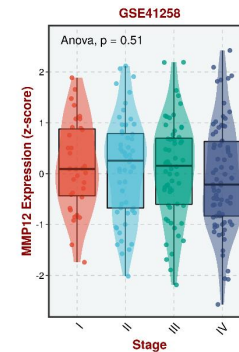
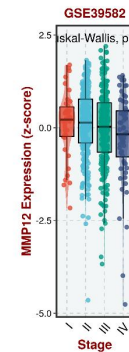
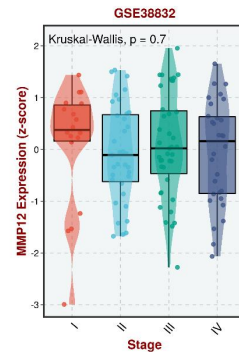
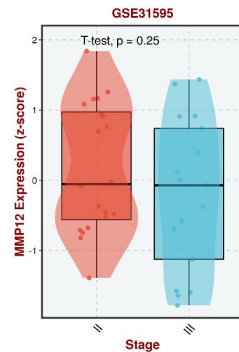
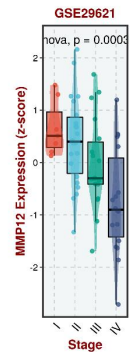
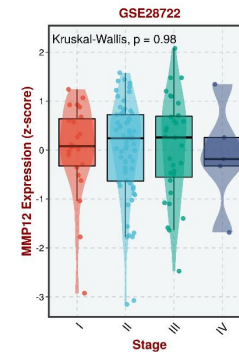
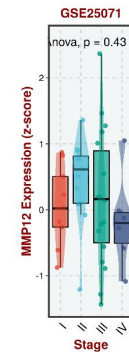
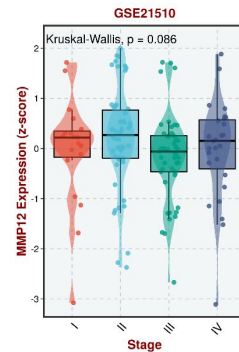
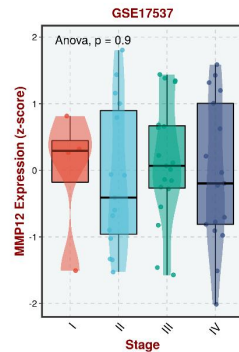
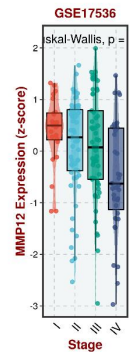
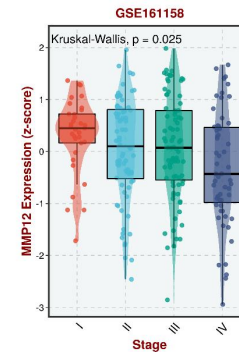
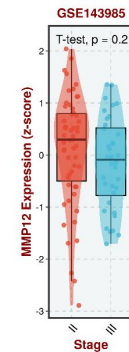
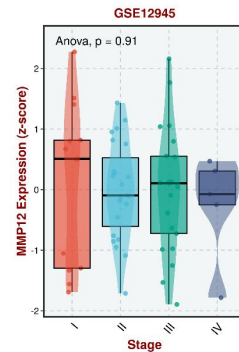
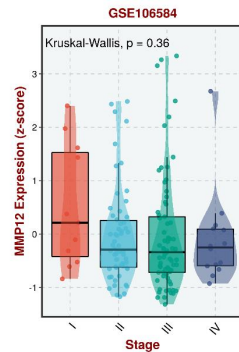
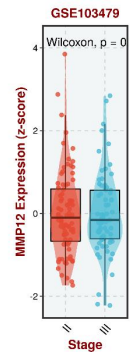


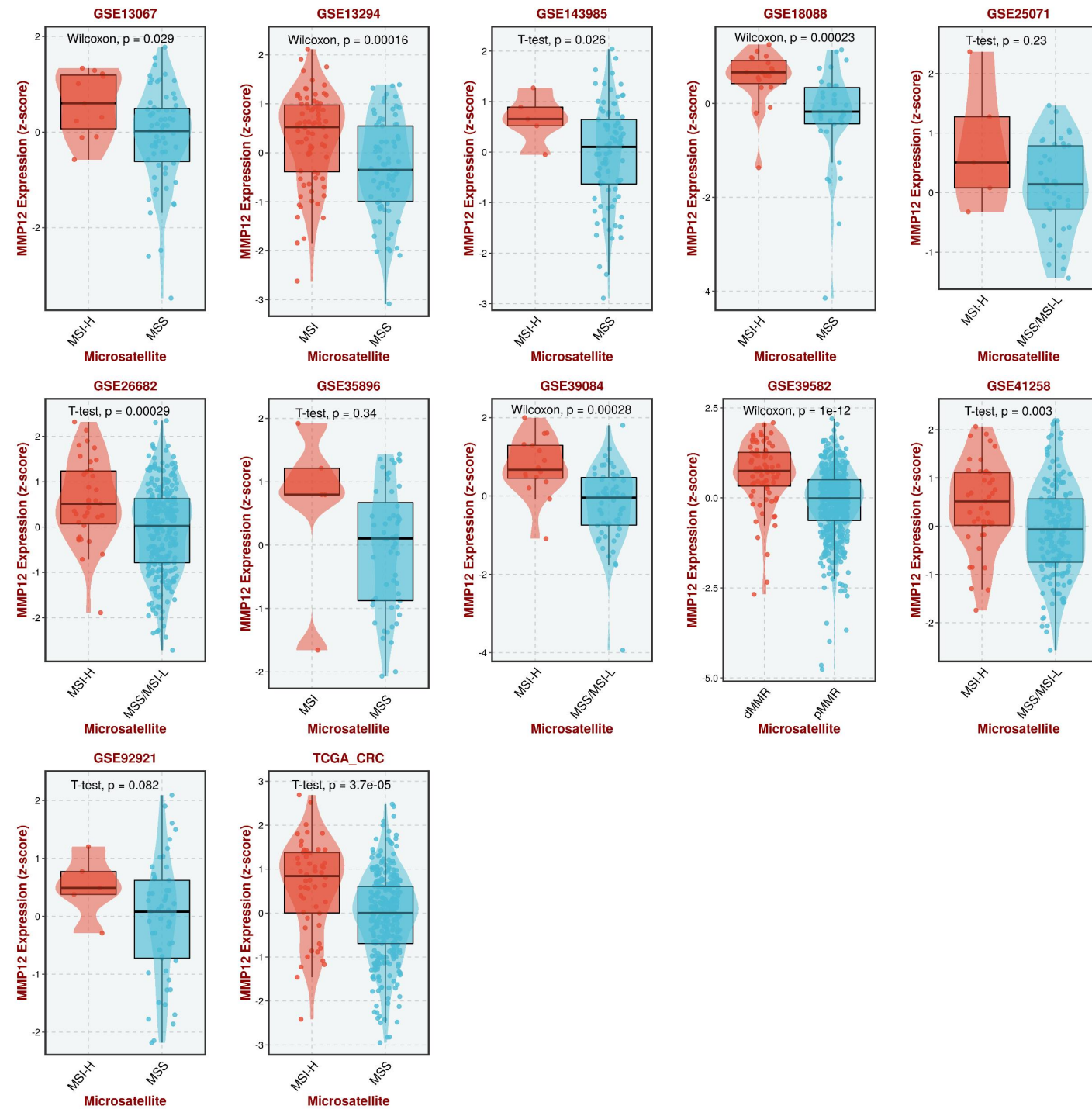
验证集验证取交集



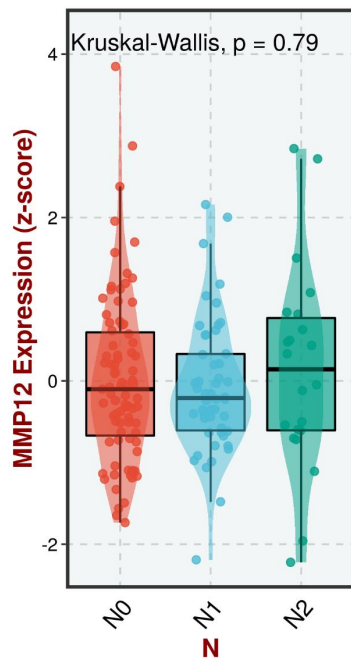
动物模型验证



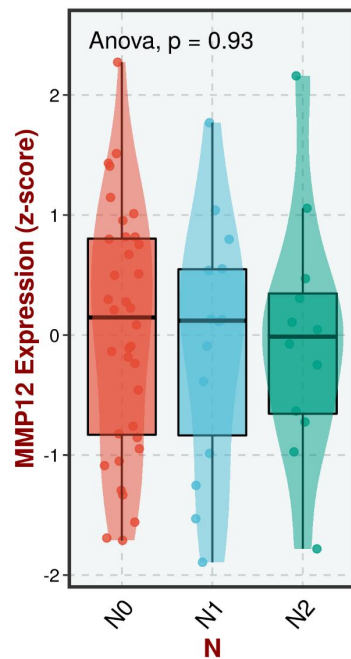




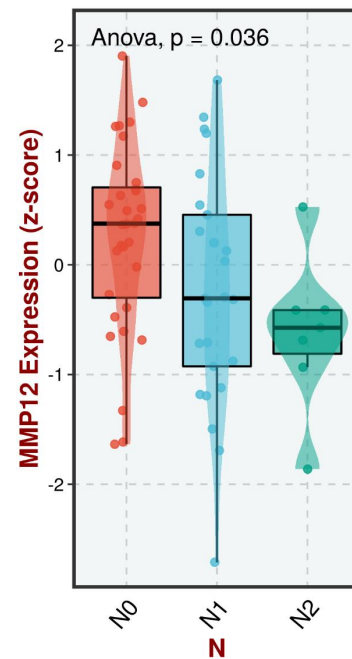
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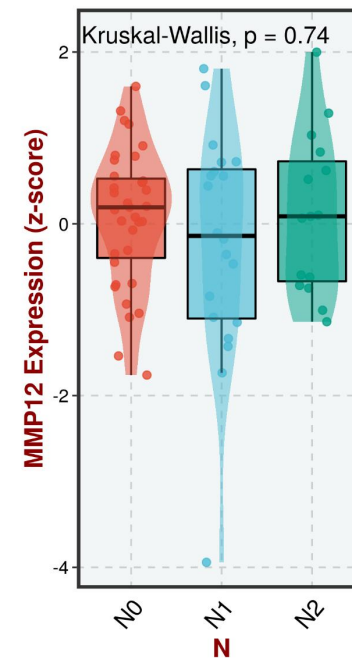
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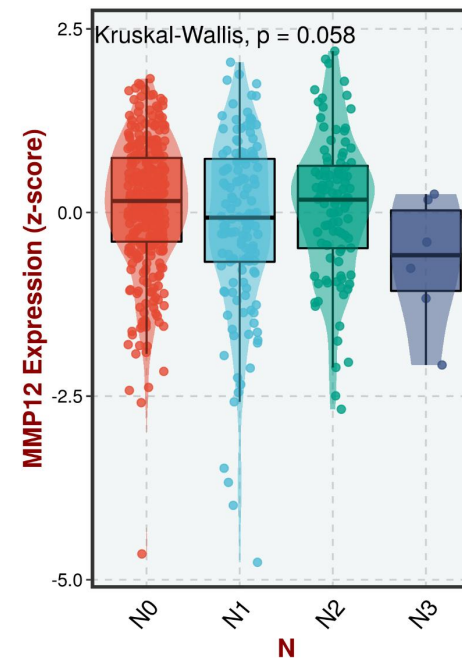
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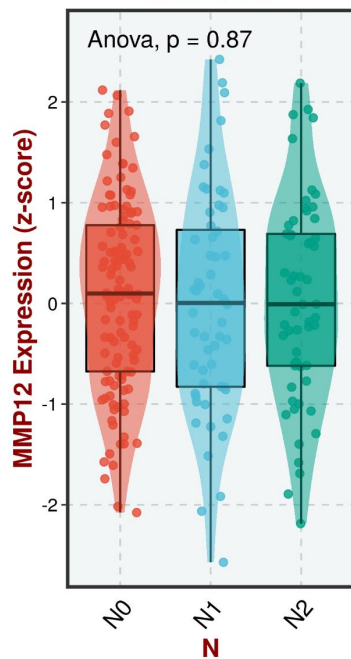
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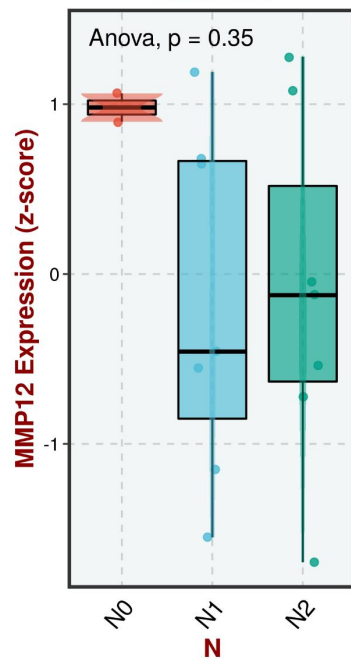
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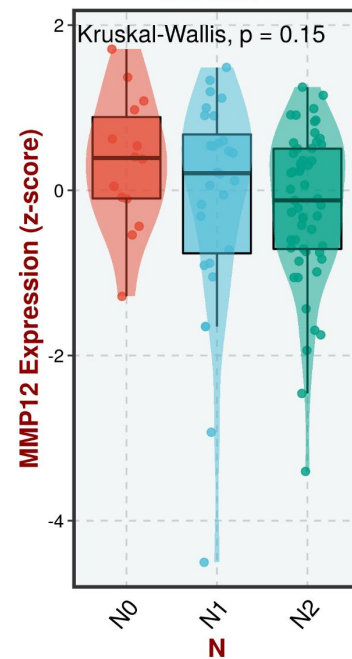
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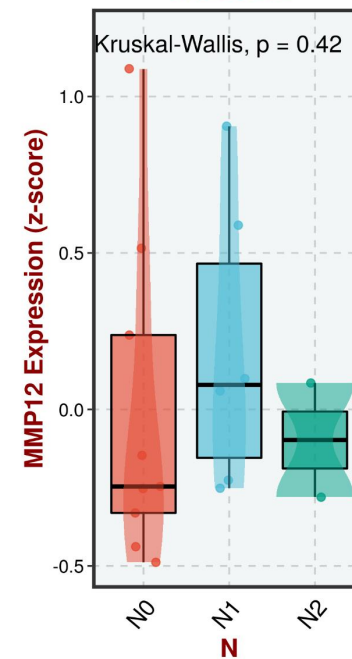
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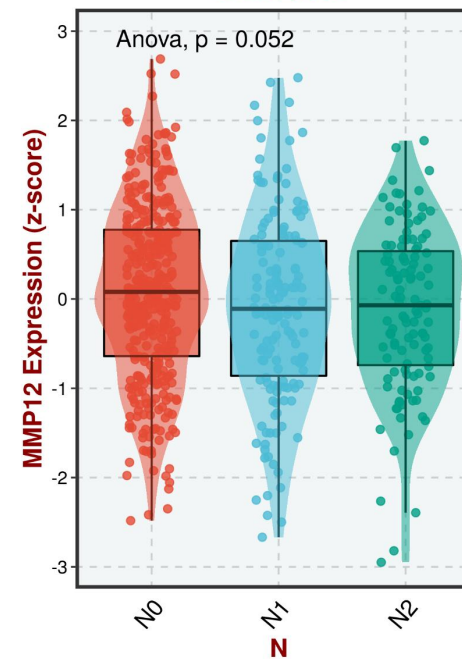
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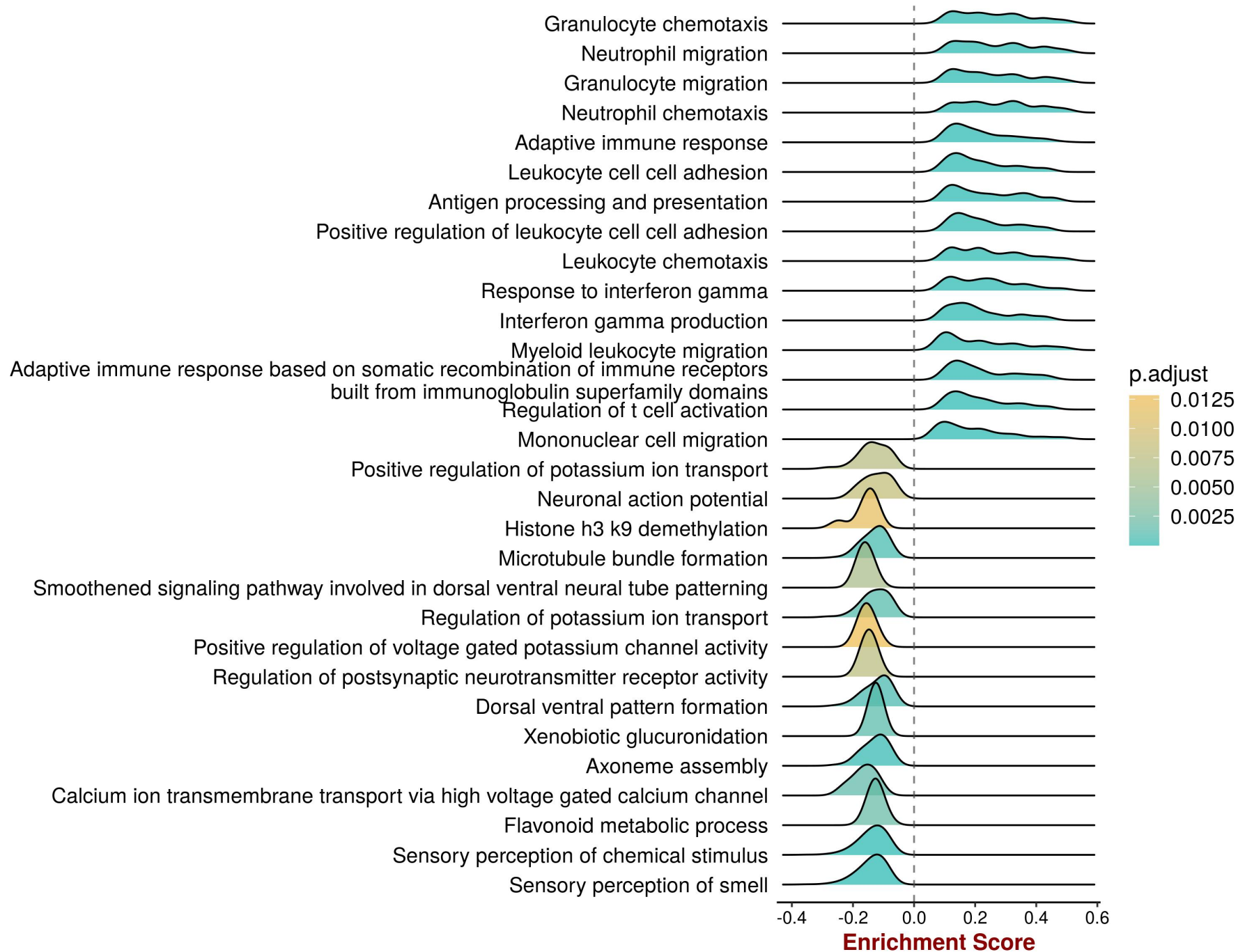
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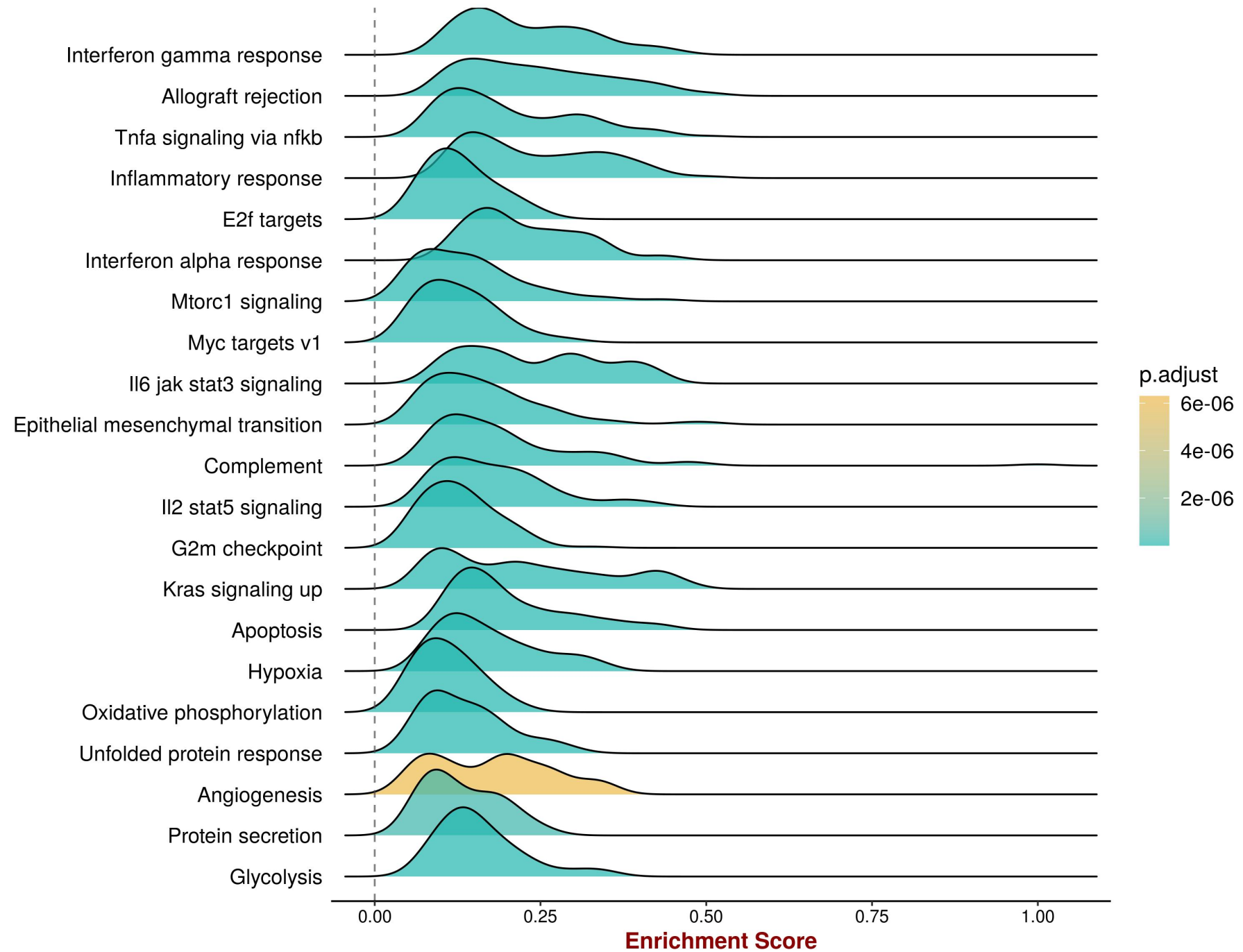
TCGA_CRC



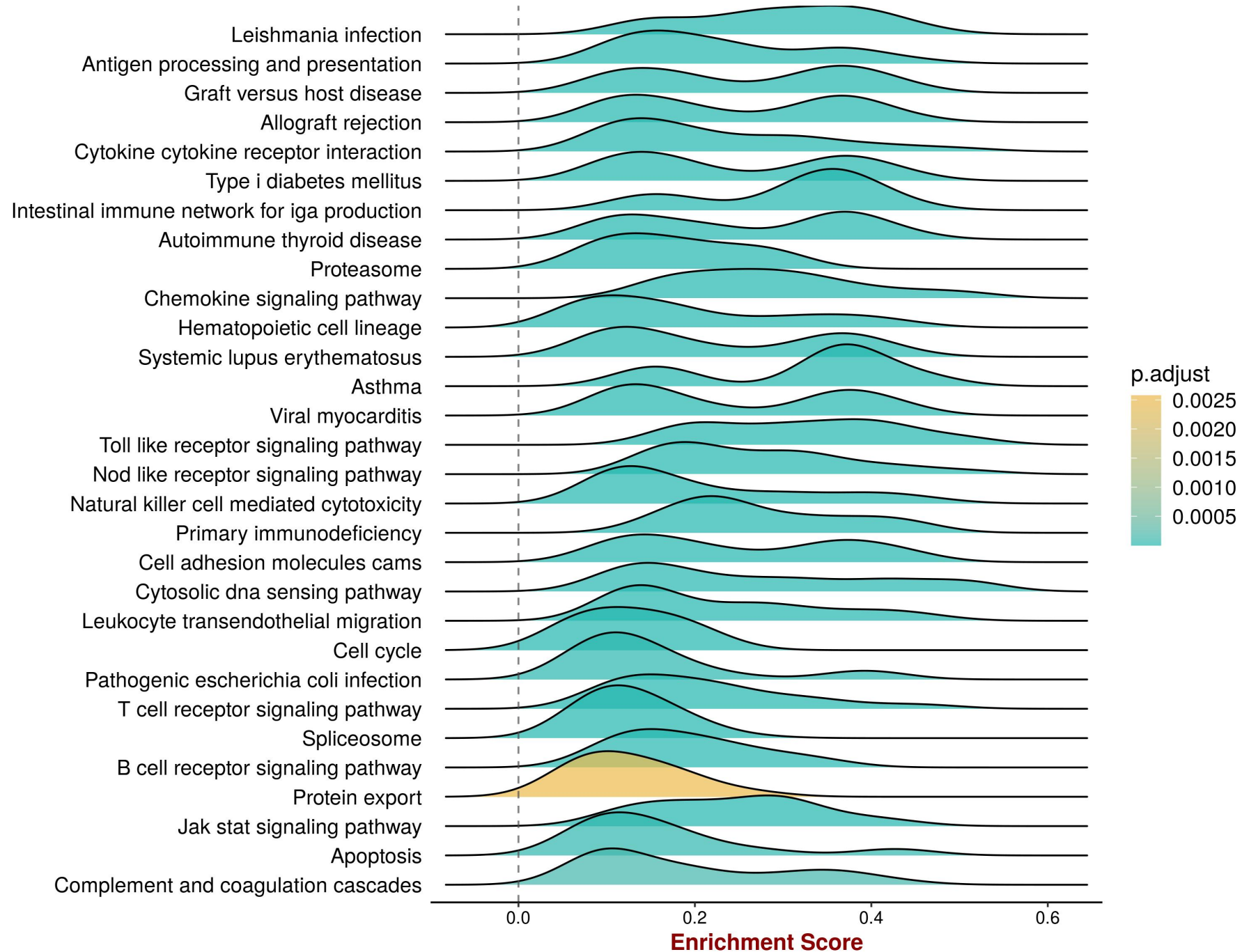
GSEA-GO Analysis

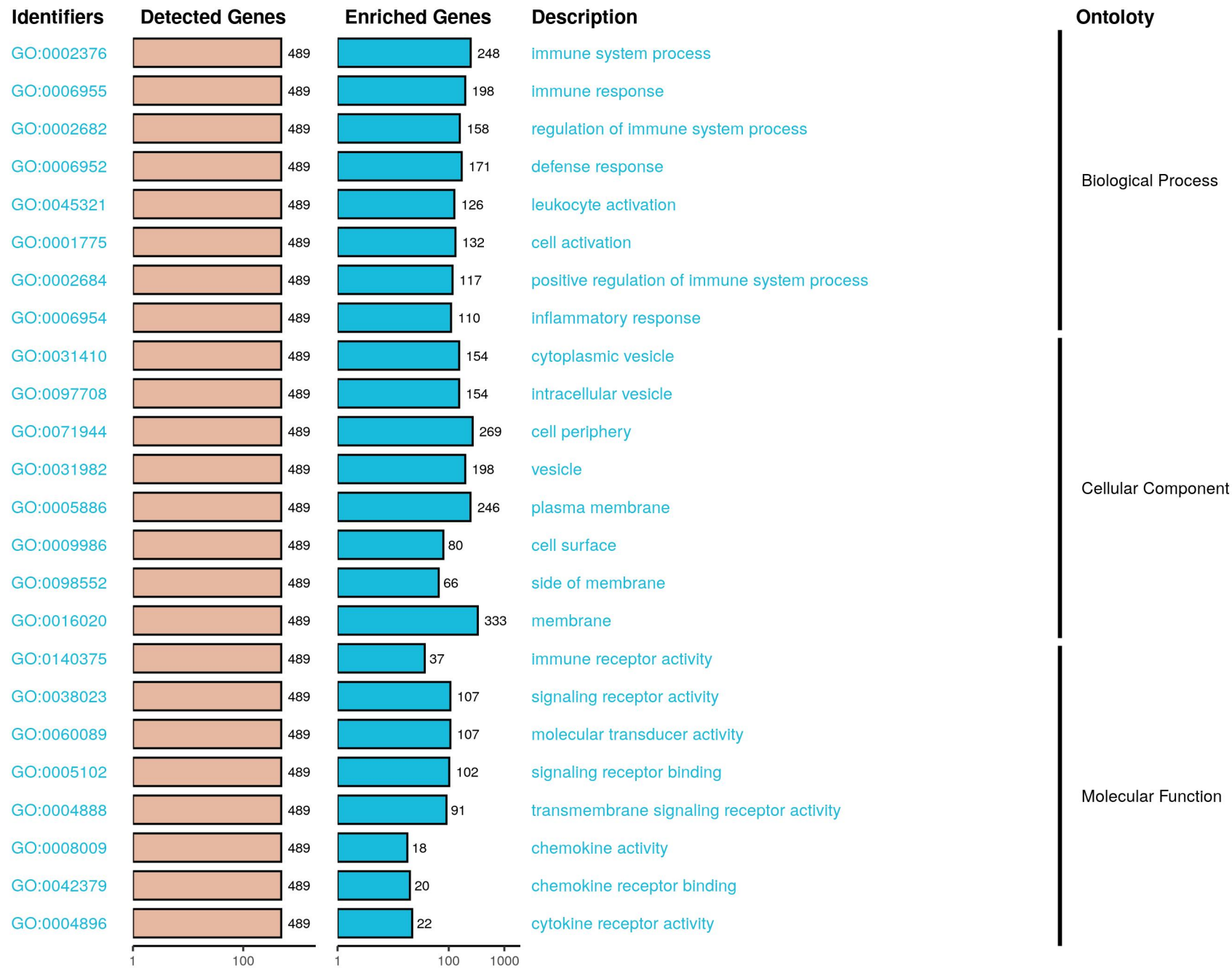


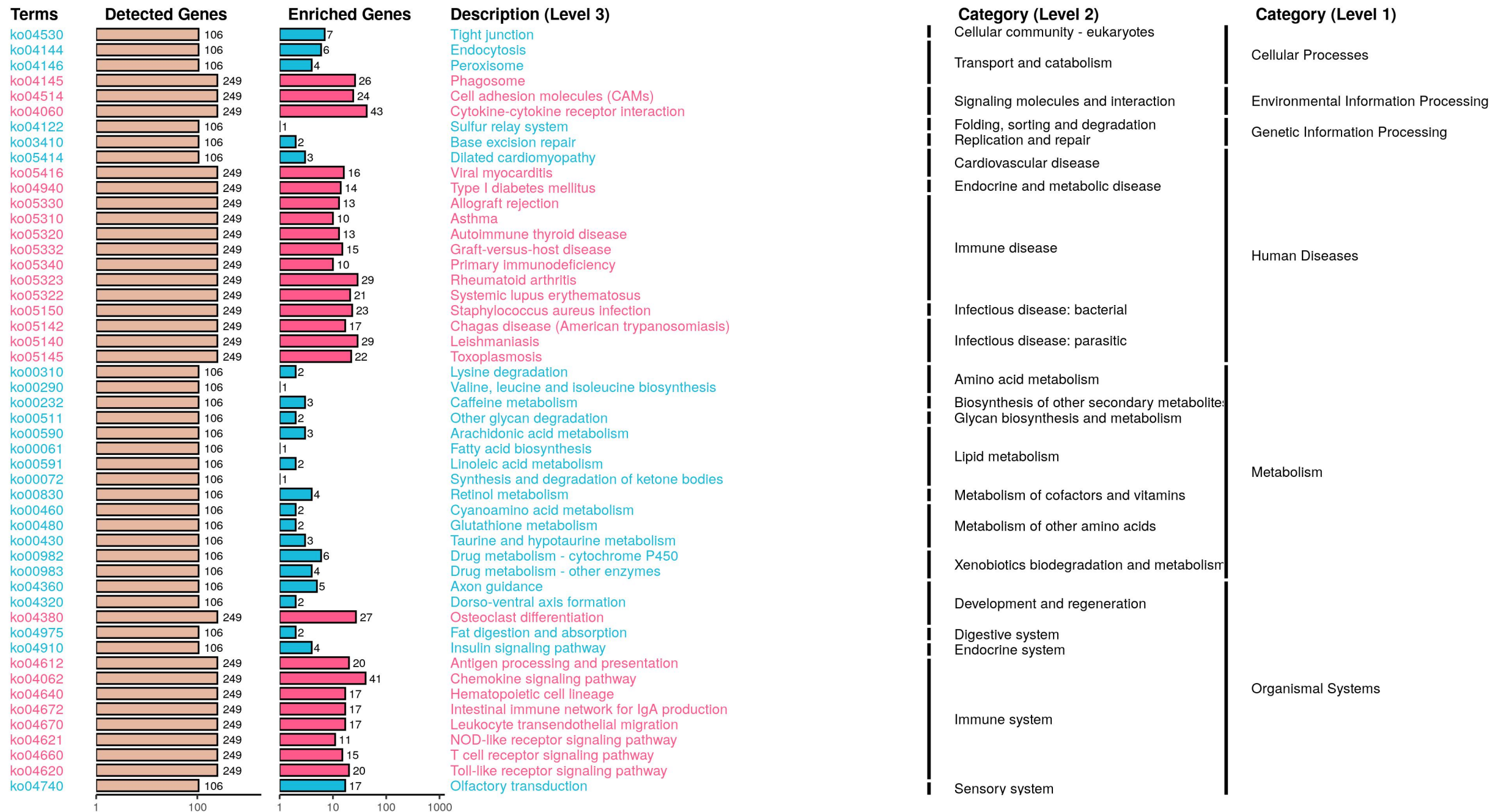
GSEA-Hallmark Analysis

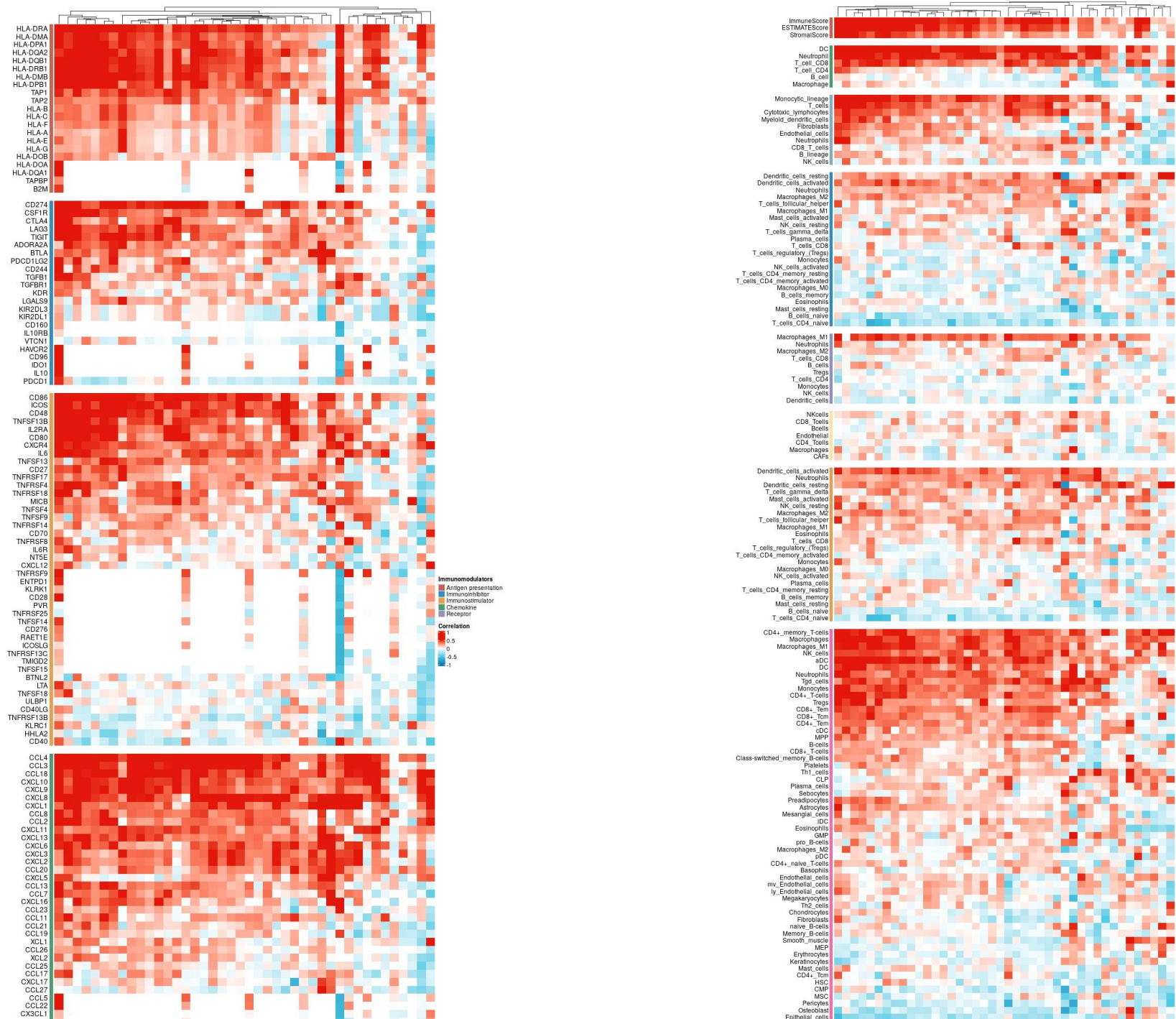


GSEA-KEGG Analysis

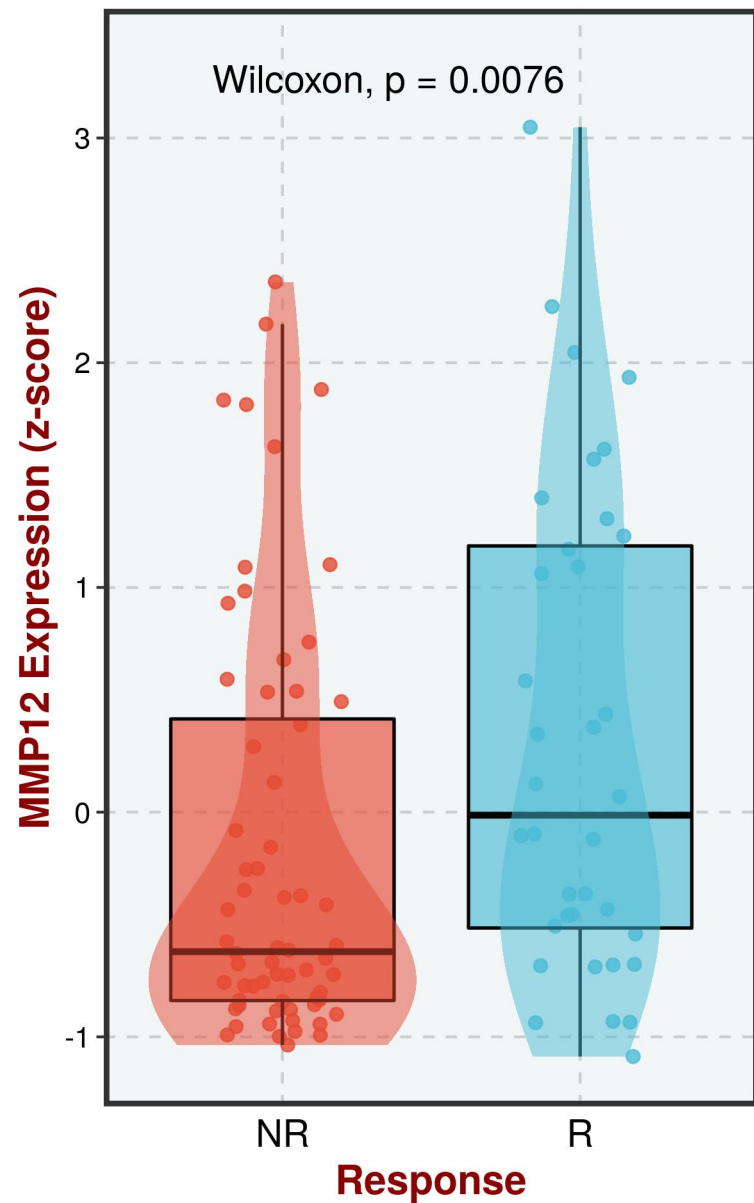




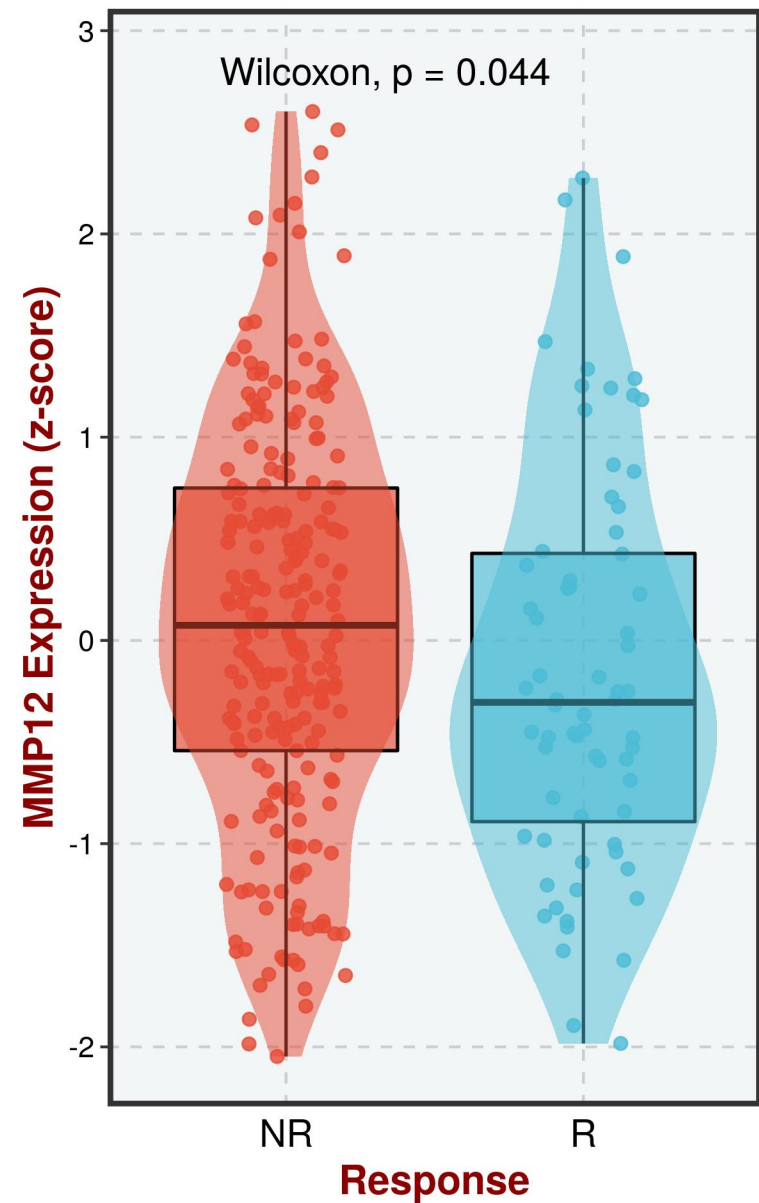


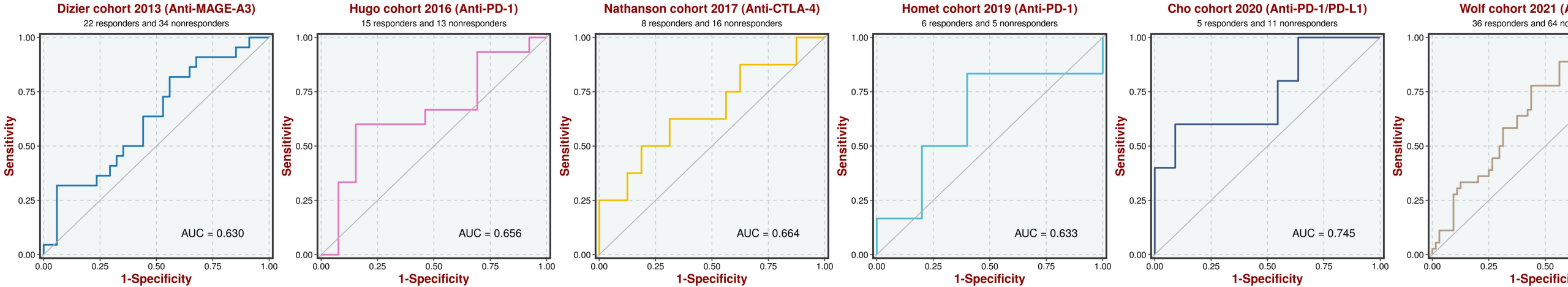


Wolf cohort 2021
Anti-PD-L1

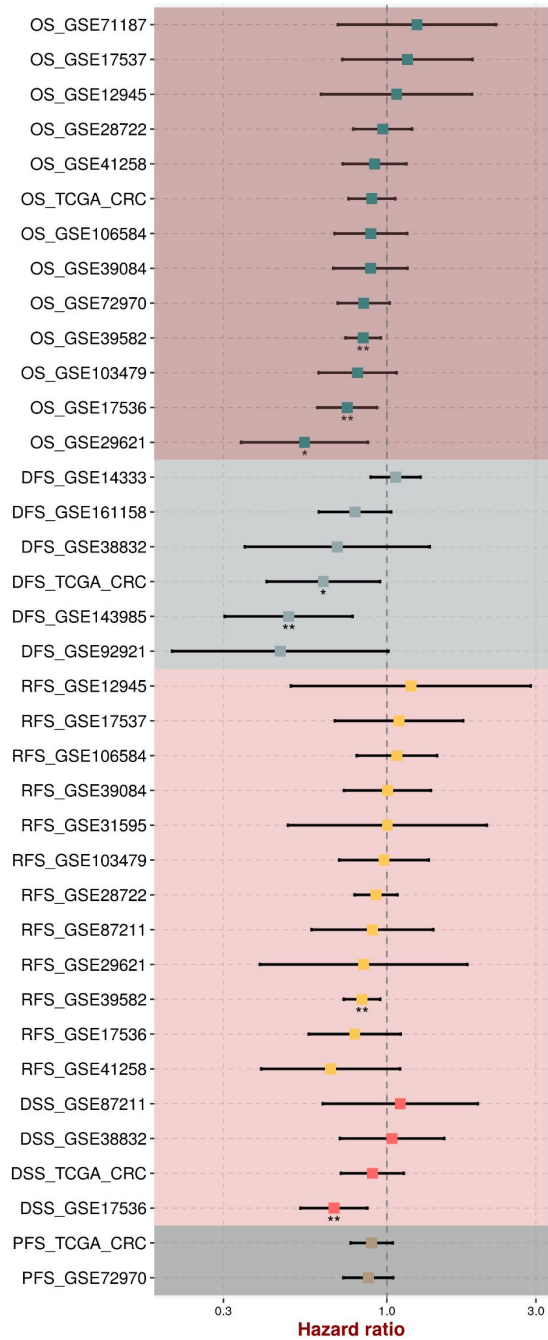


IMvigor210 cohort 2018
Anti-PD-L1

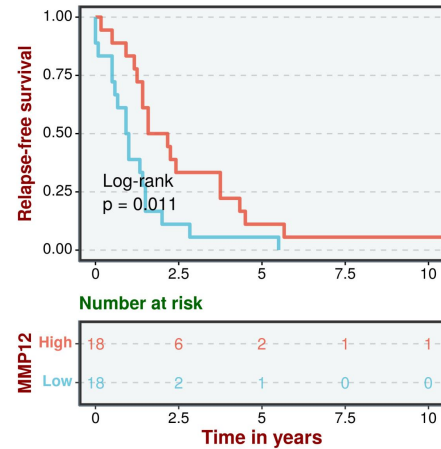




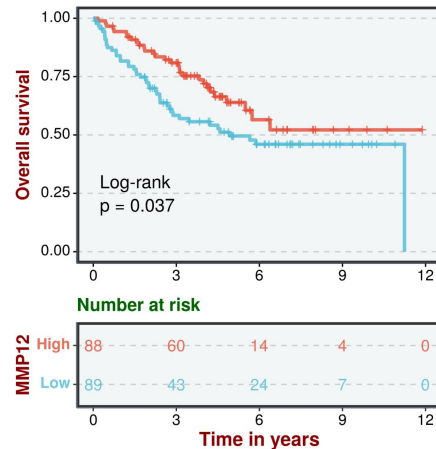
Cox regression analysis



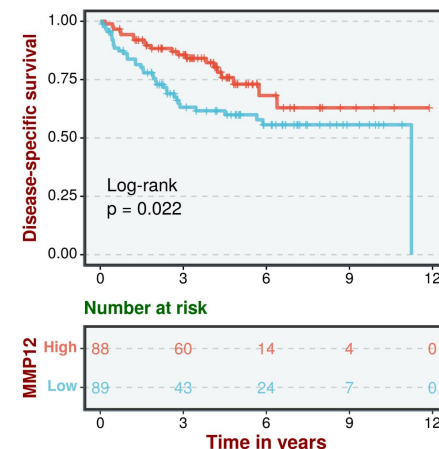
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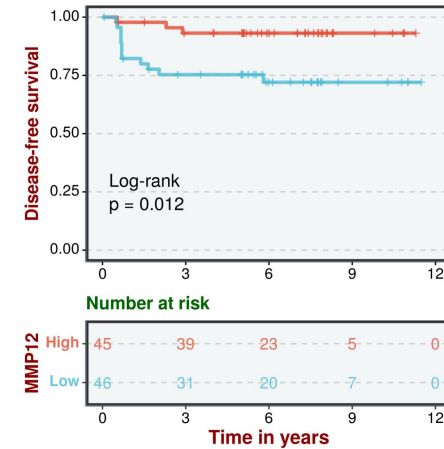
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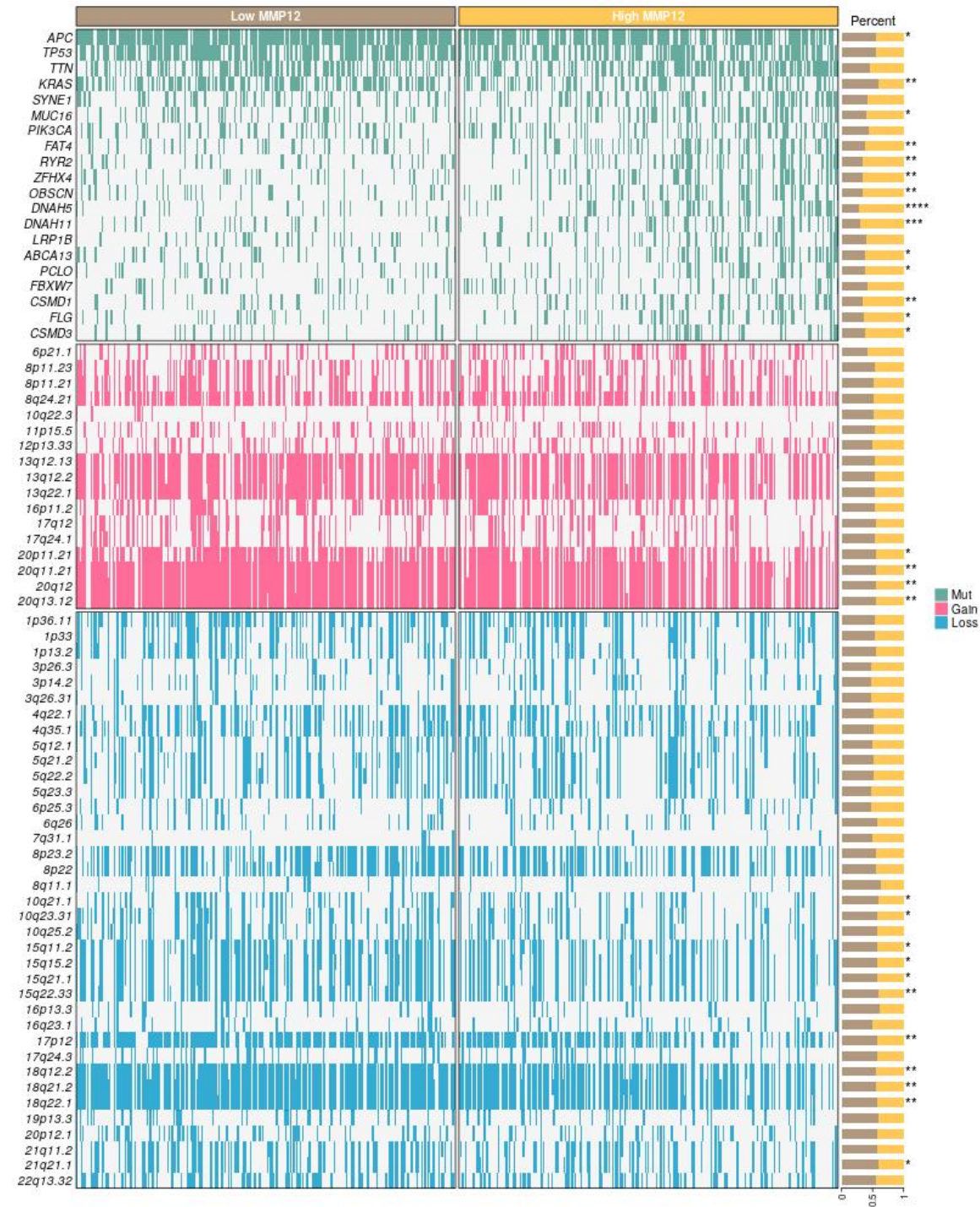
GSE17536



GSE143985



■ OS ■ DFS ■ RFS ■ DSS ■ PFS



分类	描述	基因举例
1. 转录调控与表观遗传	控制基因的开启/关闭，或通过修饰染色质来影响基因表达。	ETV6, HOXB7, ZNF195, BCOR, MSH2 (DNA错配修复，也影响基因组稳定性)
2. 代谢与酶活性	参与细胞内的各种生物化学反应，包括蛋白质修饰、核苷酸代谢等。	PFKF (糖酵解), PNP (嘌呤核苷代谢), CPD (羧肽酶), DUSP4 (磷酸酶), LYN (激酶)
3. 信号转导	在细胞内外传递信号，调控细胞生长、分化、存活等过程。	CRKL (衔接蛋白), LYN (激酶), DUSP4 (信号通路负调控), ADIPOR1 (脂联素受体), ITGA2 (整合素，细胞粘附与信号)
4. RNA加工与翻译	参与信使RNA的剪接、加工以及蛋白质的合成过程。	RBM25 (RNA结合蛋白，参与剪接), SNRPD1 (剪接体小核核糖核蛋白), EIF5B (翻译起始因子), DNAJC10 (内质网翻译相关伴侣)
5. 细胞结构与运动	构成细胞骨架，参与细胞形态维持和细胞内运输。	MYO1B (肌球蛋白，马达蛋白), DCTN4 (动力蛋白激活蛋白复合体组分), EPB41L1 (连接细胞骨架与质膜)
6. 蛋白合成、折叠与降解	负责蛋白质的正确折叠、组装以及在蛋白酶体中的降解。	HSP90B1 (内质网分子伴侣), DNAJC10 (分子伴侣), PSMC3 (蛋白酶体调控亚基), ANAPC13 (后期促进复合物，靶向蛋白降解)
7. 免疫相关	主要组织相容性复合体基因，在免疫应答中起核心作用。	HLA-A, HLA-C, BTN3A3
8. 其他/功能多样	功能独特或跨多个类别，如细胞分泌、凋亡、DNA复制等。	SCAMP1 (分泌载体膜蛋白), CASP4 (半胱天冬酶，参与炎症和凋亡), MMS19 (参与DNA修复和代谢), PLEKHA1 (参与细胞连接)